Paleomagnetism of the Hopewell Group, New Brunswick

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Samples from 15 sites in the Shepody (11) and Enragé (4) formations (Late Mississippian and Early Pennsylvanian red sediments) in New Brunswick, Canada, were subjected to thermal demagnetization. At all but two sites, the fossil magnetization appears to be composed of two nearly opposed magnetizations. The sites fall naturally into two groups: group 1 with scattered directions and group 2 with well-grouped directions. At the 8 sites of groups 1, the two magnetizations (owing to different but overlapping stability spectra) can be detected but not separated. It is shown that at 5 of the 7 sites of group 2 both magnetizations were acquired before folding. The folding was Early Pennsylvanian, indicating that the fossil magnetization was acquired in the early history of the rock and partly before and partly after a field reversal. It is suggested that the magnetization is wholly or partly diagenetic in origin. The group 2 results (mean 357°, −19°; $k = 19$; pole 34°N, 119°E) are in close agreement with previously published results from the Maringouin formation of the Hopewell group. Comparison of the Hopewell group mean direction (178°, +20°; $k = 32$; pole 34°N, 118°E) with that of Late Pennsylvanian formations indicates that an apparent polar wandering (10°–15°) took place between Early Pennsylvanian and Late Pennsylvanian-Early Permian times. Direction analysis of the two magnetizations reveals no evidence of an inclination error in these sediments.

INTRODUCTION

The term 'Hopewell group' has been applied to the beds deposited after Windsor time (Mississippian) and before Boss Point time (Pennsylvanian). In southeastern New Brunswick, the Hopewell group is divided from the bottom up into three formations: Maringouin, Shepody, and Enragé. The first two are believed to be Mississippian in age. According to *Bell* [1944], some plant remains collected from the Shepody indicate 'a very early Namurian assemblage.' The age of the Enragé is given by *Gussow* [1953] as Pennsylvanian and 'not older than Namurian or younger than Westphalian A.' Directions reported earlier from the Maringouin formation [*Roy and Robertson*, 1968] differed from those in Late Carboniferous and Early Permian beds suggesting that an apparent polar movement took place in Late Mississippian or Early Pennsylvanian time. This was the first North American evidence of an apparent polar displacement (~14°) during the Carboniferous, as has been suggested from Australian [*Irving*, 1966] and African [*McElhinny and Opdyke*, 1968] results. To determine the time and the magnitude of this displacement, the younger Shepody and Enragé beds were sampled.

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There are good exposures of Hopewell beds in the Harvey-Hopewell area and in the Maringouin peninsula. The Maringouin anticline cuts across the peninsula, the north limb dipping steeply (~85°) to the north and the south limb dipping to the south (~40°). The folding is attributed to salt tectonics; the age of the anticline is post Boss Point and pre-Pictou [*Gussow*, 1953] and therefore equivalent to the Westphalian B of Europe and the Cumberland group of the Maritimes. These beds were therefore folded not long (<25 m.y.) after deposition, so that it is possible to put a limit on the time of magnetization.

Block samples or field cores from the Shepody formation were collected at 11 sites (S) and from the Enragé formation at 4 sites (E) (Figure 1 and the appendix). In most cases, two specimens were cut from each sample or core. Sample mean directions were used as the basic data in the computations (Table 1).

RESULTS

Thirty-six pilot specimens were heated to successively higher temperature (up to 650°C) and cooled in a zero field. These pilot runs failed to detect a stable magnetization with a direction common to all sites. The direction changes during demagnetization were always

TABLE 1. Analysis of Intensities and Directions before (M_N) and after (M_T) Thermal Cleaning at Temperature T

			M_N (NRM)						M_T (Cleaned)							
Site	N_{sp}	N_{sa}	M_i	D, deg	I, deg	R	k	α_{95}	T(°C)	M_i	D, deg	I, deg	R	k	α_{95}	Lat. (°N), Long. (°E)
									Group 1							
S1	10	5	4.5	046,	+18	3.56	3		650	3.5	054,	−30	3.48	3		
S5	4	2	7.6	087,	+36	0.36	1		650	3.9	329,	−69	1.81	5		
S6	4	2	4.1	142,	+02	1.48	2		650	2.4	093,	+01	1.78	5		
S9	9	5	4.6	170,	+20	4.51	8		650	3.6	188,	+17	3.60	3		
S10	6	3	10.9	133,	+09	2.92	25		650	2.5	344,	−14	2.60	5		
E1	10	9	4.2	171,	+70	8.91	63		650	0.9	165,	+62	6.28	3		
E2	2	2	8.0	339,	−26	1.64	3		550	2.7	339,	+00	1.62	3		
E4	17	9	7.0	176,	+20	8.93	116		600	1.2	037,	−43	5.28	4		
			6.0†							2.3†						
									Group 2							
S2	12	6	6.1	020,	+35	4.90	5	35	650	5.9	010,	−16	5.72	18	16	36, 105
S3	6	2	3.4	336,	+61	1.95	20	58	550	3.2	347,	+08	1.99	177	19	48, 135
S4	5	3	14.7	188,	+09	2.96	47	18	550	14.5	190,	+06	2.98	75	14	40, 102
S7	5	5	2.9	166,	−06	3.83	3	49	550	5.4	343,	−34	4.86	28	15	24, 133
S8	6	5	5.2	131,	−32	3.70	3	52	650	11.9	004,	−23	4.89	35	11	32, 111
S11	11	6	3.6	144,	+01	3.83	4	41	650	5.3	346,	−31	5.90	50	09	30, 134
*		*6*	*5.1*†							*6.7*†	*357*,	*−19*	*5.69*	*16*	*17*	*34, 119*
E3	6	3	10.8	176,	+20	2.98	118	11	650	5.3	179,	+22	2.99	303	07	32, 117
‡		**7**	**5.6**†							**6.5**†	**357,**	**−19**	**6.68**	**19**	**14**	**34, 119** (7, 15)§

Notes

N_{sp}, N_{sa} = number of specimens, samples.
M_i = intensity $\times 10^{-6}$ gauss.
k = precision estimate [*Fisher*, 1953].
R = resultant vector.
α_{95} = radius of confidence at $p = 0.05$.
D is declination east of true north.
I is inclination positive downward.
* Italicized characters give the values for the Shepody formation.
† Logarithmic mean of the site means.
‡ Boldfaced characters give the values for group 2 using site as unit weight.
§ Semi-axis (δp, δm) of the ellipse of confidence ($p = 0.05$).

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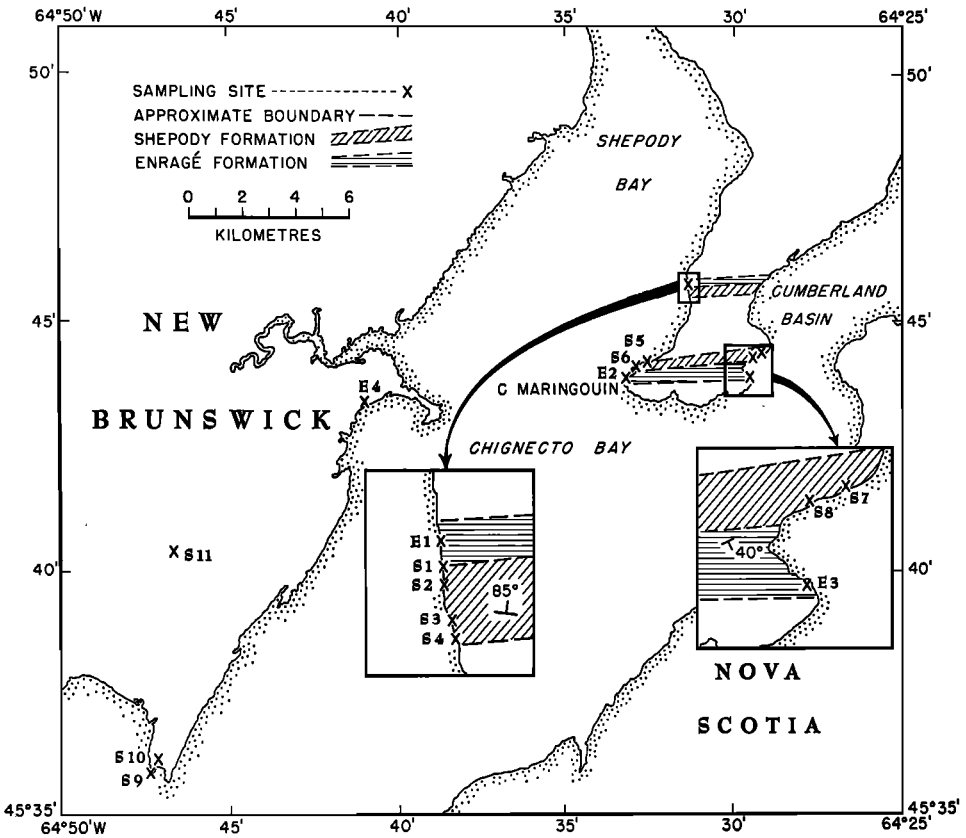


Fig. 1. Locality map showing the sampling sites.

large up to 550°C except at two sites. In the 550°–650°C range, the changes were sometimes small and sometimes large. No clear relationship could be found between these changes and the form of the intensity decay curves (Figure 2). It was therefore decided to perform a blanket demagnetization of all specimens using at least two of the following temperature steps: 500°, 550°, 600°, or 650°C. In Table 1 the temperature at which the minimum within-site dispersion was observed is given together with the mean site direction and associated statistics. The sites fall into two groups according to their within-site precision; group 1 with low precision ($k \leq 5$) and group 2 with higher precision ($18 \leq k \leq 303$) corresponding to standard deviations of greater or less than 20°, respectively.

Group 1. There are eight sites in this group. Direction changes after demagnetization were usually large (Figure 3) indicating that the natural remanent magnetizations (NRM's) are composed of two or more magnetizations. In many instances (for example, sites S10 and E4, Figure 3) the direction of these two magnetizations can be seen to be almost anti-parallel. The large dispersion after thermal cleaning suggests that neither of these magnetizations has been completely eliminated. Furthermore, in many instances, the directions of two specimens from the same core sometimes differ by more than 80° (site S10, Figure 3). It is then concluded that the site mean directions of this group cannot be used for the determination of the ancient field.

Group 2. The mean sample directions of the seven sites of this group are given in Figure 4. Very large direction changes occurred during demagnetization at sites S7, S8, and S11, the mean site direction changing by 141°(S7), 106°(S8) and 144°(S11) and the individual sample directions (Figure 5*a*) changing by al-

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most 180°. The intensity ratios M_T/M_N (Table 1) at these three sites are respectively 1.9, 2.3, and 1.5. These observations are consistent with the hypothesis of two oppositely directed magnetizations of different blocking-temperature spectra. For instance, as the intensity vector diagram (Figure 5*b*) shows, the NRM direction may diverge widely (54°) from an hypothetical axis drawn along the cleaned direction (*A*-650°C); yet, this direction could still be the resultant of two almost oppositely directed magnetizations. The true angle of Ψ shown here is likely to be smaller than 202° ($202° > \Psi > 180°$). The length of *A after* cleaning at 650°C which has been subtracted from the NRM (to find the direction of the magnetization removed by cleaning) is probably smaller than the original length of *A before* cleaning. A longer *A* prior to cleaning would reduce Ψ, and therefore the 22° misalignment must be considered an upper limit. At sites S2 and S3, the direction changes upon demagnetization were less but were still more than 50°. The intensities remain approximately constant through all the demagnetization steps (Figure 2*a*; Table 1). These results indicate the presence of two magnetizations whose directions diverge by more than 90°. Most of the direction changes at sites S2, S3, S7, S8, and S11 occur below 550°C (Figure 5*a*) although a slight increase in precision is sometimes obtained (S2, S8, and S11) by a further cleaning at 650°C. The remaining two sites (S4 and E3) are different because they do not show appreciable direction changes upon demagneti-

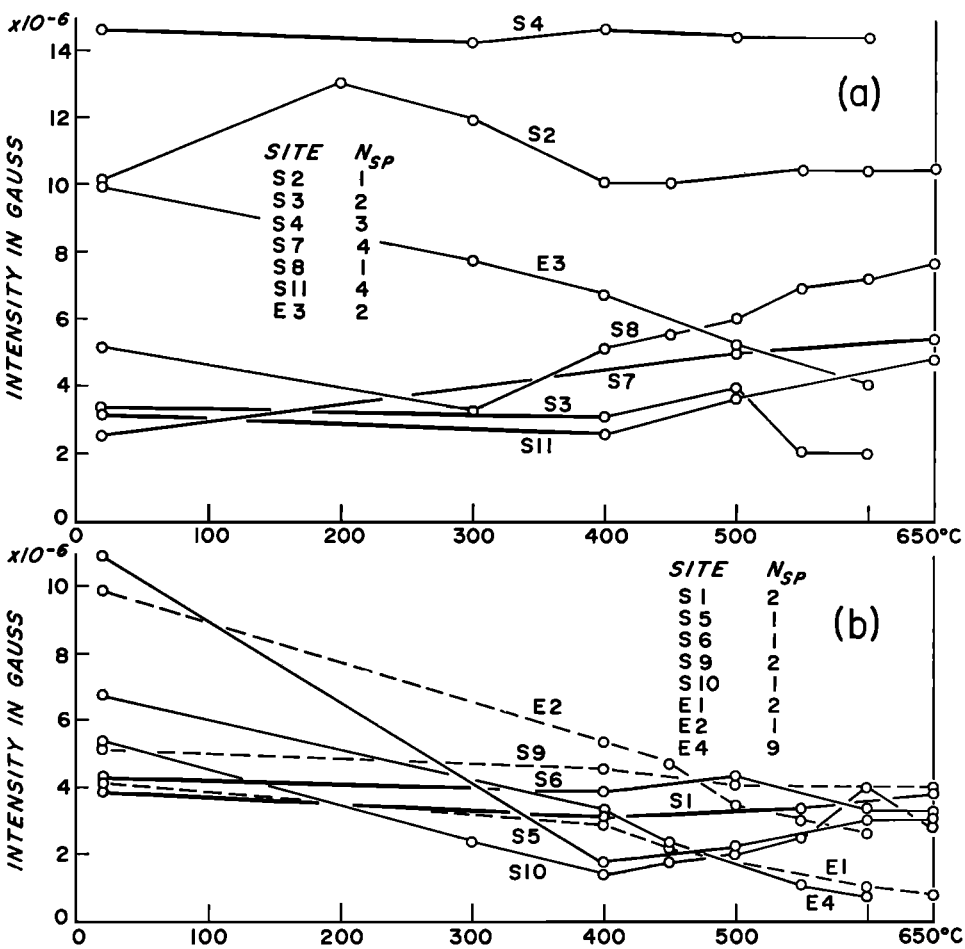


Fig. 2. Thermal decay of intensities of specimens from each site; N_{sp} indicates the number of specimens contributing to each curve.

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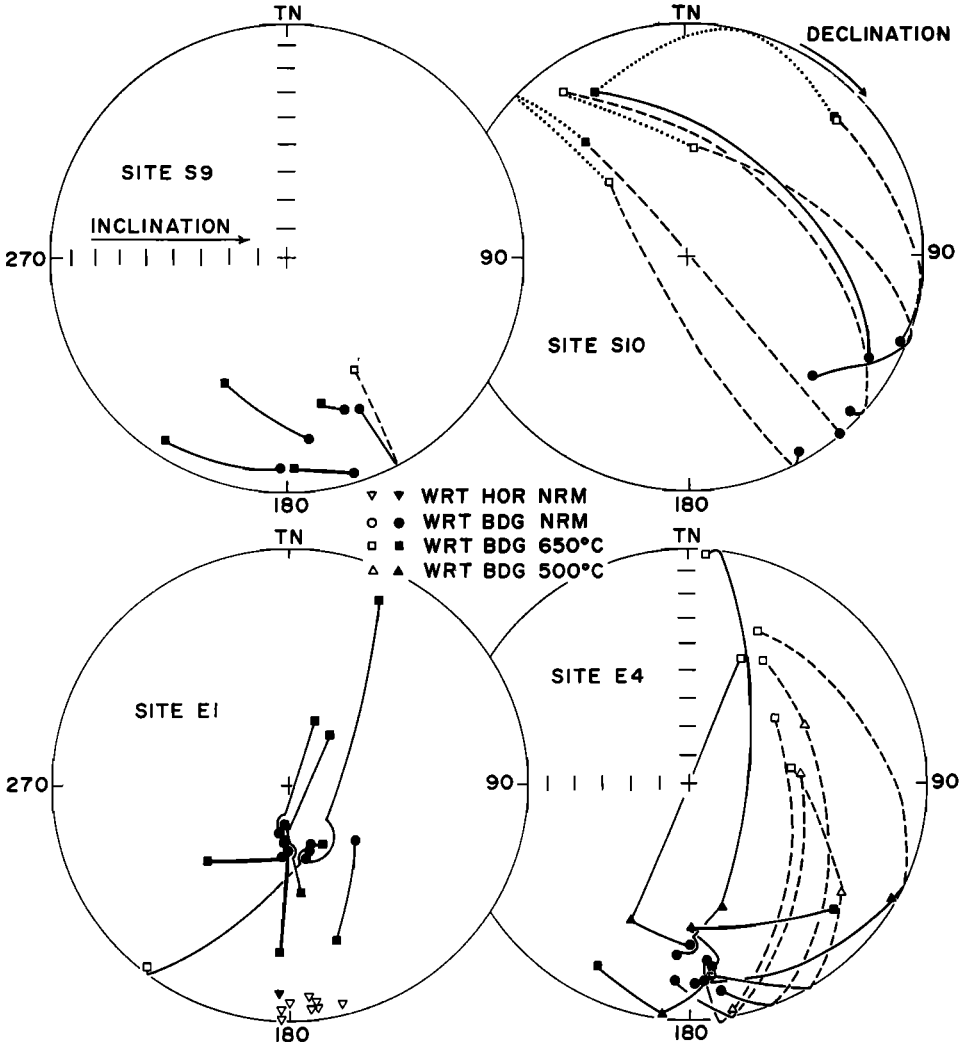


Fig. 3. Directions of magnetization before (M_N) and after (M_T) demagnetization at T°C. Dotted line joins two specimens from the same core and shows the stability variations from specimen to specimen. Equal area polar projection. Closed (open) symbols indicate positive (negative) inclinations.

zation. Therefore most of the magnetization appears to have been acquired in a field of constant polarity.

Fold test. Without correction for folding (Figure 6), the site mean directions (group 2) are scattered ($N = 7$; $R = 2.92$; $k_1 = 2$). After correction, the precision is improved ($R = 6.68$; $k_2 = 19$). The ratio k_2/k_1 is much larger than the ratio calculated at the 99% confidence limit [*McElhinny*, 1964] indicating that the magnetization *remaining* after cleaning was acquired before folding.

Origin of Magnetization: Reliability of Directions as Indicators of Ancient Field

It has been suggested [*Roy and Robertson*, 1968] that the Maringouin sediments acquired part or all of their magnetization over a considerable period of time during diagenesis. Hence, if a reversal of the field occurs during this period of time, the NRM would be composed of two almost oppositely directed magnetizations. This idea is applied to the group 2

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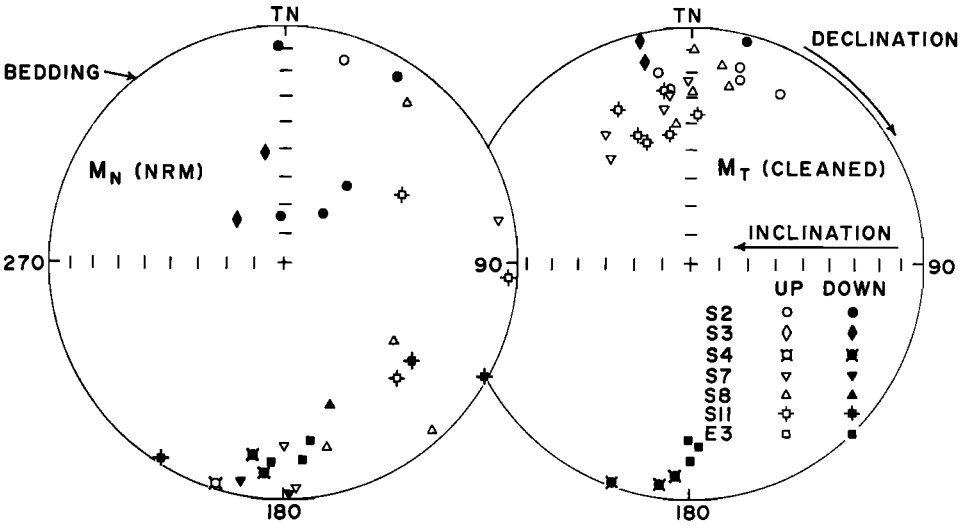


Fig. 4. Mean directions of all samples from group 2 before (M_N) and after (M_T) thermal cleaning at the temperature (T) give in Table 1. The sites are identified by symbols. Equal area polar projection.

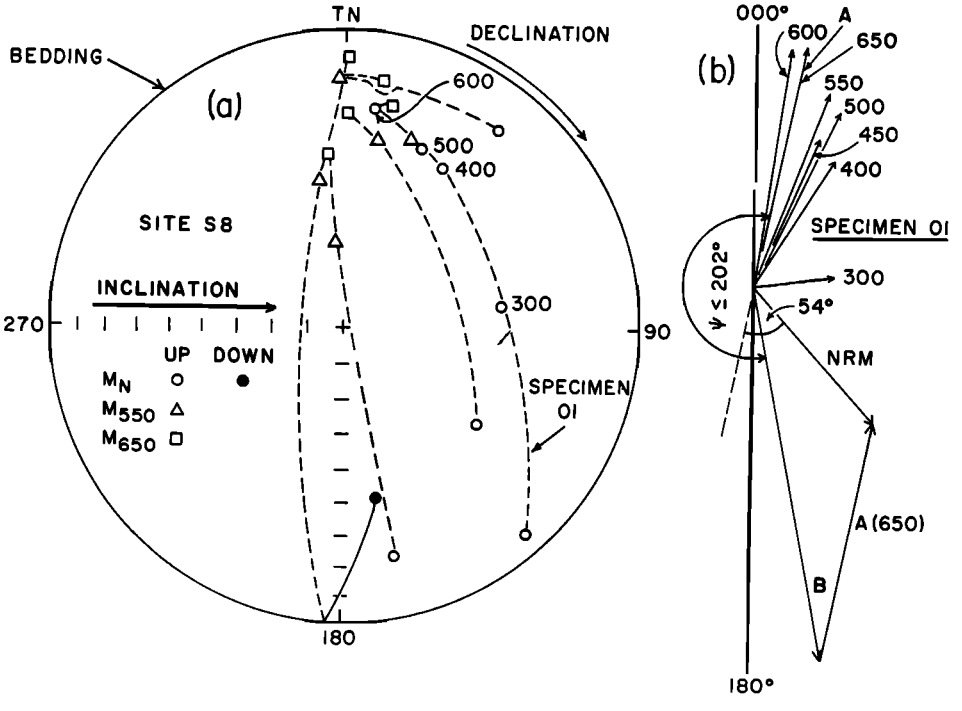


Fig. 5. Sample direction changes after thermal cleaning. Equal area polar projection is given in *a*. Horizontal intensity vectors of specimen 01 are given in *b*. Vertical component of intensity increases from −0.26 (NRM) to −3.1 (400°C) $\times 10^{-6}$ and thereafter remains approximately constant. Substraction of $A(650)$ from NRM indicates upper limit of deviation (22°) from anti-parallelism of A and B.

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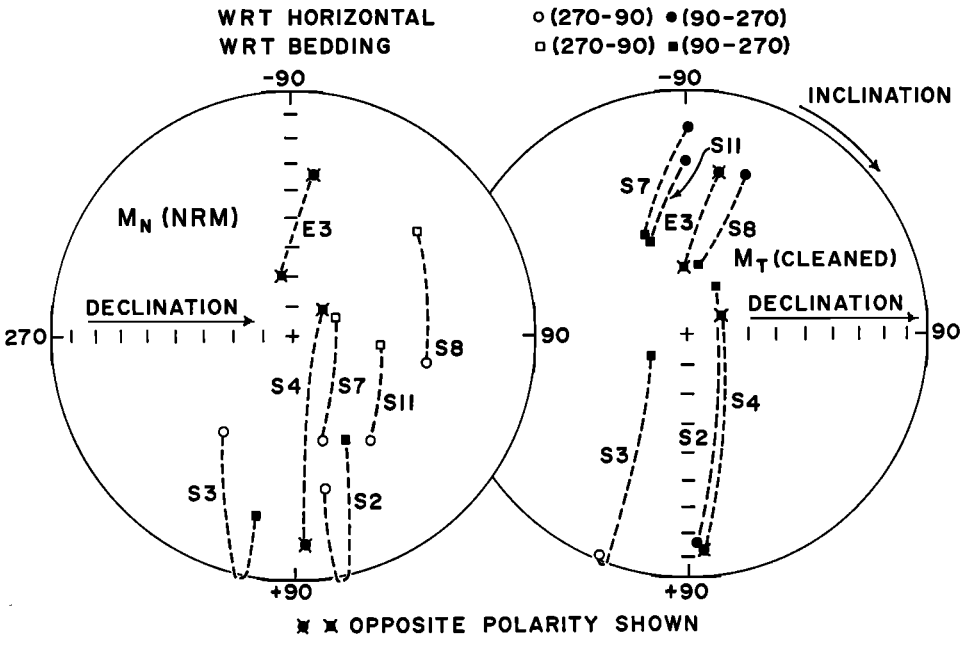


Fig. 6. Fold test for group 2 sites before and after thermal cleaning. Equal area equatorial projection.

data using the site mean directions *after correction for folding* (Figure 7).

If both magnetizations were acquired prior to folding, the direction of the magnetization removed by cleaning must lie on the great circle passing through the NRM and cleaned directions. If the magnetization removed at each site was acquired in the same field, the direction of that field will be at the intersection of all such great circles. A circle embracing all intersections of the great circles has a radius of 10° only, and its center is very close to the mean direction obtained after cleaning; this indicates that both magnetizations (that eliminated by cleaning and that remaining after cleaning), although of opposite polarity, were acquired along the same axis (within the limits set by secular variation). Furthermore, the fold test shows that both magnetizations predate the folding. It is therefore suggested that the NRM's of sites S2, S3, S7, S8, and S11 are composed of two magnetizations acquired: one before and the other after a field reversal. One or both of these magnetizations are diagenetic in origin. One cannot establish from the present data whether, in general, the more stable of the two (that remaining after cleaning) was acquired first (before reversal) or vice versa. Chemical demagnetization work, discussed below, suggests that the first magnetization is the more stable. The uncertainty, however, is unimportant for the determination of the ancient field since both were acquired before folding. Therefore, either of them *if isolated* can be used for this purpose.

In general, the resultant vector of two opposed magnetizations (acquired one before and the other after a field reversal) will not be directed along the mean axis of the ambient field owing to secular variation of that field and other perturbations. Therefore, in principle, it is necessary to remove one magnetization to determine the direction of the ancient field. This can be achieved by thermal cleaning if the blocking temperature spectra do not overlap extensively. In fact, the spectra do overlap (*Roy and Robertson*, [1968] and this paper), the degree of overlap varying widely. Consequently, the effectiveness of thermal cleaning will also vary from specimen to specimen, and the reliability of the results should be considered accordingly.

In the group 2 results, most of the direction changes at sites S2, S3, S7, S8, and S11 occurred below 550°C (Figure 5*a*). This indicates that most of the stability spectrum of the southerly

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directed magnetization belongs to a lower temperature range than that of the northerly magnetization. Also, the large increase in the within-site precision shows that the southerly directed magnetization has been greatly reduced in almost every sample. However, in some samples from sites S2, S8, and S11, small direction changes occur at high temperatures, so that the southerly directed magnetization may not be completely eliminated at 650°C. Therefore, although the individual site directions of some of these sites must be close to the direction of the ambient field, they are not necessarily parallel to it. Hence, the dispersion among these five sites may be due, in part, to incomplete removal of the southerly directed magnetization. The effect on the over-all mean direction of these five sites (and also of course of the seven sites of group 2) is not serious because of the convergence effect (Figure 7); three sites are moving through the upper hemisphere and the other two through the lower hemisphere, so that the error tends to average out. The presence of a northerly directed magnetization at sites E3 and S4 after cleaning cannot be ruled out. However, the lack of direction changes during demagnetization shows that, if present, its intensity is much smaller than that of the southerly directed magnetization and that its effect on the mean direction is small. We therefore consider that the group 2 mean direction (357°, −19°; 177°, +19°) is a fair estimate of the mean direction of the ambient field in the Maritimes during Late Mississippian-Early Pennsylvanian time.

In the group 1 results, although most within-site dispersions increase after demagnetization (Table 1), the direction changes at the sample level are not random; there is a tendency for the directions to move toward the mean direction obtained from group 2. The directions of sites E4 and S10 (Figure 3) lie along a great circle passing through two points that are close to 177°, +19°; 357°, −19° but that differ from anti-parallelism by a few degrees (≈15°). This could arise from the secular variation effects (see *Roy and Robertson* [1968]). A study of intensity and direction changes with or without correction for folding suggests that at all the group 1 sites except at site E1 both magnetizations were acquired before folding; i.e., the

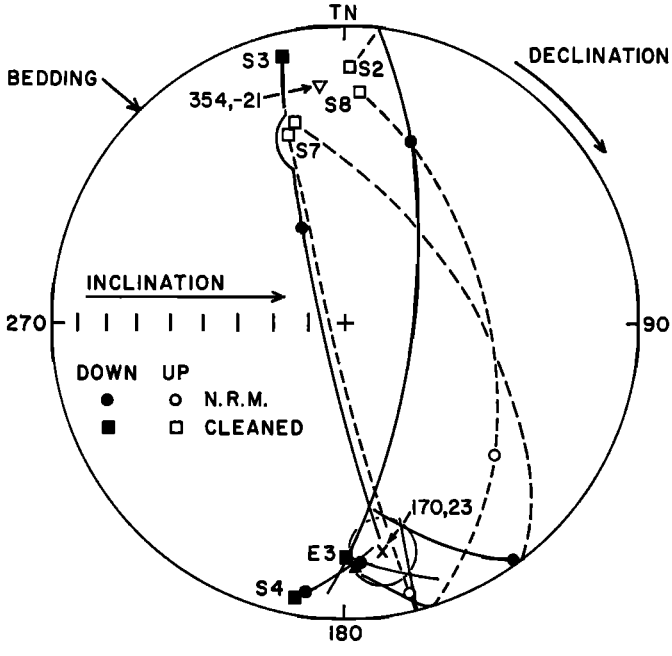


Fig. 7. Group 2 sites. Great circles passing through the NRM and cleaned directions give estimated direction (X) of removed magnetization. Closed upright triangle is group 2 mean direction (Table 1). Open inverted triangle is mean direction of the five northerly directed sites. Radius of small circle = 10°.

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changes can be explained in terms of two opposed magnetizations if both were acquired before folding but not if one or both were acquired after folding. It is therefore probable that the diagenetic process was not confined to the group 2 sites but was common to all sites, site E1 excepted; the explanation for the observed differences between groups seems to be that in group 2 the blocking temperature spectra are separate (or almost separate) but in group 1 they overlap extensively and cannot be separated by thermal cleaning.

Although the results presented so far have shown that two magnetizations could be acquired in the early history of a sediment, it has not been possible to determine which of the two (the first or the second) was removed by thermal cleaning and whether they were of different origin or not. In this respect, two results are of interest. At site E1 (which is the youngest and therefore the closest one in time to folding), the NRM's are well grouped ($k = 63$), and their mean direction (177°, −04°) without correction for folding is close to the direction of the Carboniferous field, suggesting that a magnetization was acquired shortly after folding. Since folding occurred not long after deposition of that site (<15 m.y.), this magnetization could be (as at the other sites) diagenetic in origin. After demagnetization, the directions move away and, after correction for folding, toward the group 2 direction, indicating that a smaller magnetization was acquired before folding. In this case, then, the first magnetization is thermally more stable than the second. That some directions move northward while others move southward also suggests that a field reversal took place during magnetization.

Chemical demagnetization. Experiments using *Collinson's* [1965] techniques are at present being done on a wide variety of red sediments by one of us (JKP); the details will be reported later. Preliminary results obtained from Shepody formation specimens are consistent with those obtained under thermal cleaning. On successively longer immersion times in hydrochloric acid, a specimen from site S7 changed slowly from red to white in color (small specks of black detrital minerals are clearly visible after cleaning), and the initial southerly direction gradually moved northward as in the thermal cleaning experiments. This chemical cleaning therefore indicates that the two opposed magnetizations are of different origin, the southerly one being chemical and the northerly one a DRM. The similarity of the direction changes under chemical or thermal cleaning also indicates that the DRM is more thermally stable than the chemical magnetization.

The inclination error as a possible source of uncertainty is now considered. It has been demonstrated from laboratory experiments [e.g., *King*, 1955; *Rees*, 1961] that, on deposition, sediments have a systematic inclination error; the observed inclination is less than that of the ambient field. It has sometimes been argued that natural sediments may also be subjected to an inclination error [e.g., *Nagata*, 1961, p. 242]. On the other hand, by comparing red sediments with dolorite intrusives and basalt flows, *Opdyke* [1961] found no evidence of an inclination error in the Triassic red beds of the Newark Series. In the Hopewell sediments, if the first magnetization is a depositional DRM subject to an inclination error, this would produce a misalignment of the two opposed magnetizations along the dip angle. Figure 7 shows that the southerly magnetization of S2, S3, S7, S8, and S11 (removed by cleaning) has an estimated mean direction of 170°, +23°, and that the northerly magnetization remaining after demagnetization has a mean direction of 354°, −21°; they are then exactly opposed within the statistical error. A systematic inclination error in one of these magnetizations would have resulted in a shallower inclination (~10° at these field inclinations [*King*, 1955]) for one of these magnetizations. Furthermore, in the group 1 results, more than half of the misalignments (Figure 3) are in declination rather than inclination; in the Maringouin results, most of the misalignments were also in declination. Therefore the first magnetization is either not a depositional DRM or is a depositional DRM that is not subject to a significant inclination error.

Summary of Hopewell Results

The group 2 mean direction (357°, −19°) and pole position (34°N, 119°E) are not significantly different from those of the group 2 data of the Maringouin formation (179°, +21°; 34°N, 117°E). The eight sites of the Maringouin were all southerly directed and seemed to have acquired their magnetization in a field of

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constant polarity. The within-site precisions (k) were between 38 and 391 and comparable to the values for sites S4 and E3. Table 2 gives the Hopewell data grouped by polarity.

The standard deviation of the northerly group is larger than the standard deviation attributed to secular variation of the present field [*Creer*, 1955] at that latitude (17°) whereas the standard deviation of the southerly group is smaller. This is in accord with the hypothesis that the magnetization is, in part at least, diagenetic in origin. In the northerly group, incomplete removal of the southerly magnetization would increase the standard deviation. In the southerly group, the dispersion may be too small because the magnetization at each site has been integrated over a considerable period. If so, the site mean direction does not represent a point in time but an average of a portion of the secular variation cycle. This integration would make the secular variation appear smaller than it is.

The mean directions of the two polarity groups are not significantly different ($p = 0.05$); the mean direction of each of the three formations (Maringouin, Shepody, and Enragé) are also not significantly different. This indicates that the mean pole did not undergo any appreciable changes during magnetization of the Hopewell group and that its mean direction (358°, −20°; 178°, +20°) is a good indicator of the averaged field direction in the Maritimes between Hopewell group time (Late Mississippian-Early Pennsylvanian) and Cumberland Series time (Middle Pennsylvanian).

Several studies [*Black*, 1964; *Roy*, 1966; *Roy et al.*, 1968] of formations from the same general area and deposited shortly (within 25 m.y.) after the period in question have been reported. The difference between the pole position (41°, 130°E) averaged from those results (PEI, Pictou, and Hurley Creek beds) and the Hopewell group pole position (34°N, 118°E) indicates that an apparent polar wandering of ≈12° took place in mid-Carboniferous. An apparent polar change at that time was also observed in other continents [*Irving*, 1966; *McElhinny and Opdyke*, 1968]. The magnitude of the change (~40° in Australia; 40°–50° in Africa), however, varies from continent to continent suggesting that both polar wandering and relative movement of continents took place at that time.

Appendix

All bearings are east of true north. All samples are red sandstones.

S1. West shore of the Maringouin peninsula on north limb of fold axis; 45°36′N, 64°47′W; 5 cores. Beds dipping 84° at 005°.

S2. Stratigraphically 50 meters below S1; 6 cores. Beds dipping 90° at 005°.

S3. Stratigraphically 150 meters below S2; 2 cores. Beds dipping 83° at 009°.

S4. Stratigraphically 30 meters below S3; 3 cores. Beds dipping 83° at 009°.

S5. West shore of the Maringouin peninsula on south limb of the fold axis; 2 hand samples. Beds dipping 43° at 181°.

S6. Stratigraphically 120 meters above S5; 2 hand samples. Beds dipping 53° at 187°.

S7. East shore of the Maringouin peninsula on south limb of the fold axis; 5 cores. Bed dipping 41° at 154°.

S8. Stratigraphically 60 meters above S7; 5 cores. Beds dipping 42° at 149°.

TABLE 2. Analysis of Direction of the Hopewell Group (Maringouin-Shepody-Enragé Formation)

Polarity	N	D, deg	I, deg	R	k	θ_{63} deg	α_{95}	Pole Position			
								Lat. (°N)	Long. (°E)	δ_p	δ_m
Northerly	5	354	−21	4.74	16	20	20	33	123	11	21
Southerly	10	180	+20	9.84	55	11	07	34	116	04	07
Average	15	358	−20	14.56	32	14	07	34	118	04	07

Notes

N is number of sites.

θ_{63} is circular standard deviation.

Other abbreviations are the same as given in Table 1.

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S9. West of Cape Enragé; 45° 36′N, 64° 47′W; 5 cores. Beds dipping 48° at 126°.
S10. Stratigraphically 7 meters above S8; 3 hand samples. Beds dipping 78° at 117°.
S11. Road cut 4 km south of Midway; 45°40′N, 64°47′W; 6 cores. Beds dipping 32° at 146°.
E1. Stratigraphically 100 meters above S1; 9 cores. Beds dipping 76° at 011°.
E2. Stratigraphically 160 meters above S6; 2 hand samples. Beds dipping 45° at 170°.
E3. Stratigraphically 500 meters above S8; 3 cores. Beds dipping 39° at 155°.
E4. Coast cliff 2.5 km east of Harvey; 45°43′N, 64°41′W; 9 cores. Beds dipping 55° at 163°.

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